

RESEARCH ARTICLE SUMMARY

2D MATERIALS

2D theoretically twistable material database

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INTRODUCTION: Twisting one atomically thin crystal against another creates a large-scale interference pattern called a moiré pattern, which can strongly reshape how electrons move. This idea has already produced unusual topological states, magnetism, and superconductivity in a few celebrated systems, including twisted bilayer graphene, but the much larger family of possible two-dimensional (2D) crystals has barely been explored.

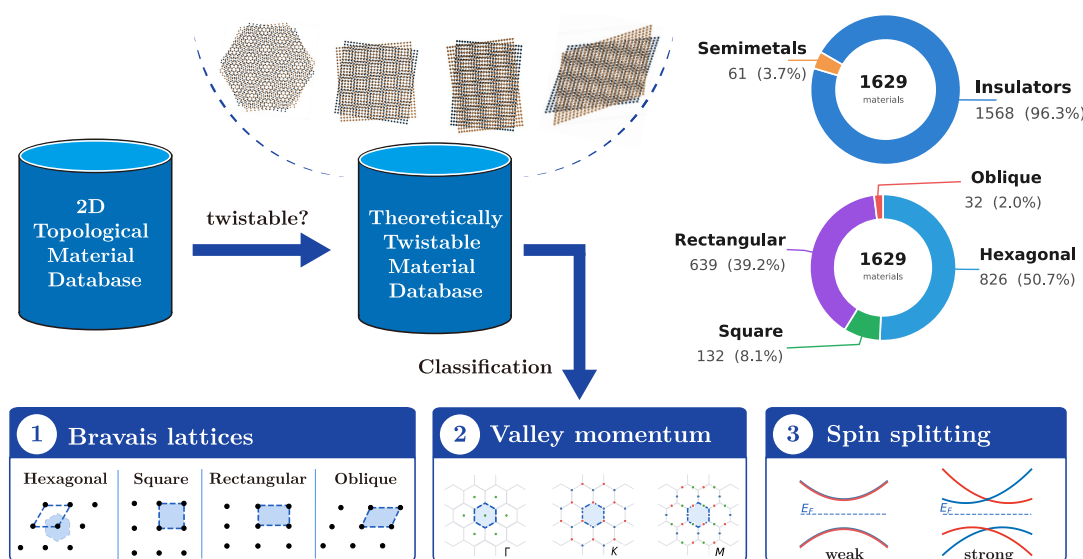
RATIONALE: The main obstacle is that not every material remains understandable after twisting. Many have overly complicated electronic structures, making reliable predictions difficult. We therefore defined “theoretically twistable” materials as 2D crystals with clean, simple electronic features near the energies of interest. We built a high-throughput search based on the Topological 2D Materials Database to find such systems among both semimetals and insulators. The screen evaluated band cleanliness, bandgap, number of nearby electronic states, crystal symmetry, and structural simplicity, and then grouped the surviving materials by lattice type, location of the electronic valleys in momentum space, and strength of spin splitting. To connect prediction to practice, we also investigated whether promising materials can be grown or exfoliated to monolayers.

RESULTS: The search identified 61 twistable semimetals and 1568 twistable insulators. These candidates greatly expand the moiré

materials landscape beyond the few material families studied so far, uncovering platforms with hexagonal, square, and rectangular lattices; with weak or strong spin splitting; and with valley arrangements that can generate honeycomb-, kagome-, or square-like plane wave lattices in momentum space. Example twisted-bilayer calculations showed isolated flat bands and other desirable electronic structures in representative materials. The study also linked theory with experiment: Several promising compounds were synthesized in bulk, multiple candidates were exfoliated into high-quality monolayers, and selected monolayers were incorporated into working twisted devices for transport and low-temperature scanning measurements. All computed properties, including effective masses, band-edge energies, and twistability metrics, were released through a public database to guide future work.

CONCLUSION: This work turns the search for moiré materials from a case-by-case effort into a systematic design strategy. By combining large-scale electronic screening with experimental feasibility, it provides a broad and practical map of 2D crystals that can host new twisted quantum matter. □

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Workflow of the database construction. Starting from a 2D topological material database, a filtering score is applied to construct a database of theoretically twistable materials, which includes both semimetals and insulators. The candidate materials in this database are further classified according to their Bravais lattices, valley splitting, and spin splitting.

2D MATERIALS

2D theoretically twistable material database

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The study of twisted two-dimensional (2D) materials, in which twisting layers create moiré superlattices, has created opportunities for investigating topological phases and strongly correlated physics. The broader potential of a seemingly infinite set of twistable 2D materials remains largely unexplored. Here, we define “theoretically twistable materials” as single- or multilayer structures that allow for the construction of simple continuum models of their moiré structures. Our high-throughput algorithm systematically searches for theoretically twistable semimetals and insulators based on the Topological 2D Materials Database (2D-TQCDB). By analyzing key electronic properties, we identify thousands of candidate materials, propose representative twistable materials, and provide examples of crystal growth and exfoliation for several of them. Our results provide a resource for future experimental and theoretical studies of moiré systems.

The discovery and study of two-dimensional (2D) materials have revolutionized condensed matter physics, opening avenues for exploring quantum phenomena in reduced dimensions. One of the most exciting developments in the field is the concept of twistronics (*1–4*), in which a small twist between layers of 2D materials leads to the formation of moiré superlattices, fundamentally altering the electronic structure and interaction landscape and giving rise to new phenomena.

Twisting introduces periodic modulations and creates a moiré pattern that can greatly affect the low-energy electronic properties. The resulting moiré bands can become extremely flat due to the enlarged moiré unit cell, quenching the kinetic energy of electrons and thus enhancing the role of electron-electron interactions. These interactions can drive a wide range of topological and correlated phases such as correlated insulating states (*5*), unconventional superconductivity (*6*), and fractional Chern insulators (*7–9*). The prototypical example is

twisted bilayer graphene (TBG) (*10*), in which a “magic angle” twist between two graphene layers leads to the emergence of flat bands (*5, 6, 11–29*). Beyond graphene, twisted transition metal dichalcogenides (TMDs), such as twisted MoTe₂ and WSe₂, have also been shown to exhibit many different types of strongly correlated phases (*30–52*).

Despite the intense focus on TBG and TMD systems, many other potential moiré systems remain unexplored. Given the vast number of 2D materials (*53–56*) with different crystal symmetries and electronic structures, there is a much larger space of twistable materials that could host rich correlated and topological physics (*57–65*). The systematic exploration of this space is crucial for identifying new systems with properties that go beyond what has been observed in TBG and TMD moiré superlattices.

Systematic progress, however, is limited by the lack of theoretical guidance. Many monolayer, bilayer, and multilayer systems have complex band structures that cannot be reduced to simple effective models, such as the Dirac cone in graphene or the quadratic band edge in TMDs. Twisting such systems produces moiré bands that are difficult to model, obscuring the prediction of “magic” angles (*10*). In TBG, such a theoretical prediction was essential for identifying the superconducting regime. We therefore define “theoretically twistable” materials as exfoliable 2D materials with clean, low-energy band structures. For metals, this requires point- or line-like Fermi surfaces amenable to effective modeling. For insulators, it requires isolated band extrema that support simple low-energy descriptions. We also impose experimentally motivated criteria such as a band gap <3 eV, because larger gaps make electrical contacting difficult. Further criteria are described below.

On the basis of these criteria, we developed a high-throughput algorithm to screen the Topological 2D Materials Database (2D-TQCDB) (*66, 67*) for theoretically twistable materials. The algorithm identifies semimetals and insulators with clean electronic structures suitable for predictive moiré engineering. Applying it, we found 61 twistable semimetals and 1568 twistable insulators, which we classified by Bravais lattice, valley type, and spin-orbit coupling (SOC) strength. For representative materials in each class, we present twisted bilayer band structures and corresponding simplified moiré continuum models. We also discuss experimental feasibility. The full dataset, including twistability metrics, effective masses, and vacuum-aligned band-edge energies, is available through 2D-TQCDB. This work expands the materials basis for twistronics and provides a practical guide for future experimental and theoretical studies of 2D moiré systems.

Classification of twistable materials

In twisted systems, the moiré potential breaks the translation symmetry of the monolayer and couples the electron states connected by the moiré reciprocal lattice vectors $\mathbf{G} \in Q = \mathbb{Z}\mathbf{b}_{M_1} + \mathbb{Z}\mathbf{b}_{M_2}$, where $\mathbf{b}_{M_i} = 2\sin\left(\frac{\theta}{2}\right)\mathbf{b}_i \times \hat{\mathbf{z}}$ are moiré reciprocal vectors defined based on the monolayer reciprocal vectors $\mathbf{b}_{1,2}$ and twist angle θ . The single-particle Hamiltonian for moiré systems takes the form

$$\mathcal{H} = \sum_{\mathbf{k}, \mathbf{Q}, \mathbf{Q}', i, j} [h_{\mathbf{Q}, \mathbf{Q}'}(\mathbf{k})]_{ij} \hat{c}_{\mathbf{k}, \mathbf{Q}, i}^\dagger \hat{c}_{\mathbf{k}, \mathbf{Q}', j} \quad (1)$$

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where $\mathbf{k}$ takes value in the first moiré Brillouin zone (BZ) and i and j are composite indices for orbital, spin, and other degrees of freedom. $\mathbf{Q}$ takes values in the moiré $\mathbf{Q}$ lattice, i.e., $\mathbf{K}_{\text{valley}} - \mathbf{Q}$, which is formed by the moiré plane wave components expanded around the valley momentum $\mathbf{K}_{\text{valley}}$. Depending on the Bravais lattice and valley types, moiré systems exhibit a diverse range of properties.

We classify twistable materials into two types. The first type includes semimetals with (approximately) zero-dimensional (0D) Fermi surfaces (FSs), such as graphene with a linear Dirac crossing. These semimetals may feature various crossing types at the Fermi level, including linear and quadratic degenerate points. The second type includes insulators with clean valence band maxima (VBM) or conduction band minima (CBM), exemplified by TMD materials. We refer to both the crossing points in semimetals and the VBM/CBM in insulators as the “twisting points,” which are not necessarily located at high-symmetry momentum points (HSPs). Each of the two types of twistable materials is further classified based on (i) the Bravais lattice, (ii) the momentum of the twisting point, and (iii) SOC splitting at the twisting point.

The four Bravais lattices corresponding to 80 layer groups (LGs) are hexagonal, square, rectangular (including centered rectangular), and oblique. The valleys are labeled according to their momenta as follows: for hexagonal lattices, Γ , M, and K; for square lattices, Γ , X, and M; and for rectangular and oblique lattices, only HSPs for simplicity. Non-HSPs apply to all lattices. The HSP labeling follows the conventions of the Bilbao Crystallographic Server (68). For SOC classification, a numerical threshold of 50 meV energy splitting near the twisting point is used to identify materials with strong SOC.

In Fig. 1, we illustrate the moiré $\mathbf{Q}$ lattices generated by different Bravais lattices and valley configurations. For a hexagonal lattice, the Γ valley gives a triangular $\mathbf{Q}$ lattice, whereas the K valleys form a honeycomb lattice with two sublattices that correspond to the time-reversal symmetry (TRS)-related valleys $K_{1,2}$, as shown in Fig. 1A. The M valleys instead generate a kagome $\mathbf{Q}$ lattice in Fig. 1B. This case was introduced in (69) for SnSe₂ and ZrS₂, which are representative materials from the current high-throughput results and can be exfoliated into single layers in experiments. They were shown to host emergent nonsymmorphic symmetries. For a square lattice, the Γ and M valleys generate square $\mathbf{Q}$ lattices, whereas the X valleys form two nested square lattices associated with the C_4 -related valleys $X_{1,2}$, as shown in Fig. 1, C and D. For a rectangular lattice, both the Γ and Y valleys generate rectangular $\mathbf{Q}$ lattices, as shown in Fig. 1E. Thus, the Bravais lattice and valley configuration determine the geometry of the moiré $\mathbf{Q}$ lattice. We have already described the nontrivial physics of twisting the M point in the hexagonal lattice with negligible SOC (69). This represents just one of the universality classes of twisted materials, with each class presented here likely to host previously unexplored physics.

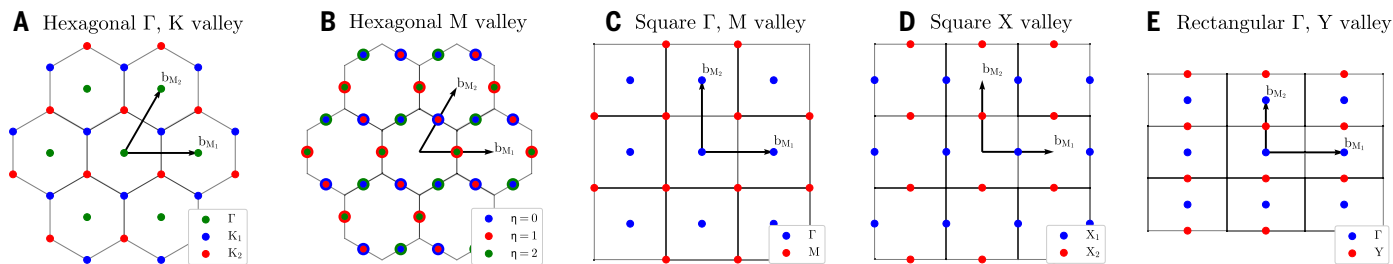


Fig. 1. Moiré $\mathbf{Q}$ lattices derived from different Bravais lattices and valleys. (A) $\mathbf{Q}$ lattices corresponding to the Γ and K valley in the hexagonal lattice. The K valley generates two sets of $\mathbf{Q}$ lattices from the two TRS-related K points, K_1 and K_2 , forming a honeycomb lattice in momentum space. (B) $\mathbf{Q}$ lattice from the M valley (69) in the hexagonal lattice, which results in a kagome lattice in momentum space, with three sublattices corresponding to the three C_3 symmetry-related M valleys labeled by $\eta = 0, 1, 2$. For each valley, solid (hollow) circles of the same color correspond to the $\mathbf{Q}$ lattice of the top (bottom) layers. (C) Γ and M valleys from the square lattice, both forming a square lattice in momentum space. (D) $\mathbf{Q}$ lattice generated by the X valley from the square lattice, forming two nested square lattices in momentum space given by the two C_4 -related valleys X_1 and X_2 . (E) $\mathbf{Q}$ lattices for the Γ and Y valleys from the rectangular lattice.

High-throughput algorithm

We now outline the high-throughput screening algorithm for identifying theoretically twistable materials, illustrated in Fig. 2. We first excluded conventional metals with an odd number of valence electrons, which generally cannot form 0D Fermi surfaces or insulating states. We then restricted the search to monolayer unit cells with at most 12 atoms and four chemical elements.

To quantify the isolation of low-energy bands, we defined a “clean energy range”: the energy window around the twisting point containing only one band without SOC or one Kramers pair with SOC. This condition ensures that the band edge admits a simple $\mathbf{k} \cdot \mathbf{p}$ description and is likely to yield clean moiré bands after twisting. The detailed criteria for twistable semimetals and insulators are given below.

For semimetals, the criteria are as follows:

1) Enforced-semimetal condition: the topological classification of the material, either with or without SOC, must be either an enforced semimetal with Fermi degeneracy (ESFD), featuring crossings at high-symmetry points, or an enforced semimetal (ES) with crossings along high-symmetry lines, thereby ensuring a symmetry-protected crossing point between valence and conduction bands. Semimetals with a small SOC gap at the crossing points are also included.

2) Density-of-states (DOS) condition: the DOS at the Fermi level E_f must be ≤ 2 states per unit cell per electron volt. This threshold takes an empirical value and aims to account for large DOS contributions from quadratic crossing points, small Fermi surface pockets from other bands, or multiple crossing points. For instance, AuTe features several crossing points at E_f , resulting in a relatively large DOS of about 1 state per unit cell per electron volt. Nevertheless, the bands near E_f remain relatively clean, making it a viable twistable candidate.

3) Clean-energy-range condition: both the VBM and CBM must exhibit a clean energy range of at least 200 meV. This energy range is on the order of the interlayer hopping in moiré systems, within which a reliable low-energy description can be achieved.

4) Band-edge condition: the highest valence band edge and lowest conduction band edge should be within 0.1 eV from E_f . Even if the semimetal is gapped by SOC or other mechanisms, the gap is typically not large enough to fully separate the conduction and valence bands. As a result, the theoretical description must account for both bands together. We also allow small FS pockets within 0.1 eV from E_f , because they may be eliminated upon twisting.

For insulators, we applied the following criteria: (i) gap condition: a global band gap between 200 meV and 3 eV; (ii) clean-energy-range condition: a clean energy range of at least 200 meV in either the highest valence band or the lowest conduction band; and (iii) DOS condition: we require the maximal DOS within 200 meV to the twisting point should be ≤ 4 states per unit cell per electron volt. This criterion

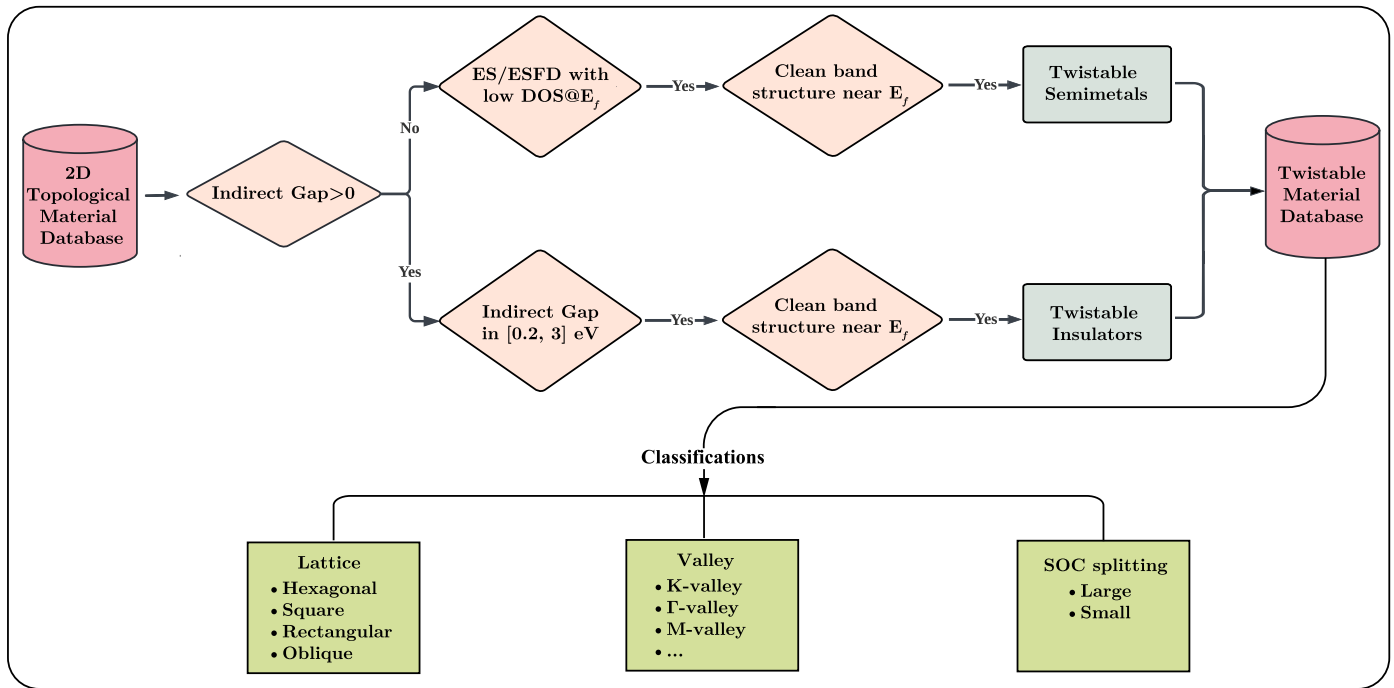


Fig. 2. High-throughput algorithm used to search for 2D twistable materials. We start from all 2D materials in the 2D-TQCDB (66). For a given material, if it has a zero indirect gap, we then determine whether it has a topological classification of ES or ESFD, a low DOS, and a clean band structure at the Fermi level. If these criteria are met, then the material is classified as a theoretically twistable semimetal with symmetry-protected degeneracy at the Fermi level. Conversely, if the material has an indirect gap within the range of [0.2, 3] eV and a clean band edge in either the valence or conduction bands, it is considered a theoretically twistable insulator. The identified twistable semimetals and insulators are further categorized based on their Bravais lattice, valley momentum, and SOC splitting strength.

also takes an empirical threshold used to exclude extremely flat bands at the twisting point, which are likely to produce spaghetti-like bands after twisting.

These conditions are applied to ensure well-isolated band structures. A maximum gap of 3 eV was chosen because a larger gap would be experimentally challenging for making electrical contacts and performing effective gating. Given that ab initio calculations generally underestimate the band gap, compounds with large gaps will be ranked with low scores defined in the following. We note that multiple, symmetry-unrelated valleys may exist within the clean energy range. In practice, these valleys are typically either energetically distinct or effectively decoupled at the single-particle level after twisting. Therefore, they are included in our algorithm.

To rank the twistable materials identified by the algorithm, we have established a scoring system to quantify their suitability. The scores for semimetals and insulators are defined as follows:

$$\begin{aligned}
 S_{\text{semimetal}} &= \frac{1}{5}(S_{\text{DOS}} + S_{\text{cleanVB/CBRange}} + S_{\text{cleanCRange}} \\
 &\quad + S_{\text{symmetry}} + S_{\text{atom number}}) \\
 S_{\text{insulator}}^{\text{VB/CB}} &= \frac{1}{4}(S_{\text{gap}} + S_{\text{cleanVB/CBRange}} \\
 &\quad + S_{\text{symmetry}} + S_{\text{atom number}})
 \end{aligned} \quad (2)$$

Here, the score for twistable insulators, i.e., $S_{\text{insulator}}^{\text{VB/CB}}$, is defined for the twisting point at VBM and CBM separately. The criteria for each component are as follows:

1) Gap score (S_{gap}): aims for an ideal indirect band gap Δ range of 1 to 1.5 eV. The score decreases linearly with deviation from this range, i.e., $S_{\text{gap}} = 1$ for $\Delta \in [1, 1.5]$ eV, $S_{\text{gap}} = 1 - |\Delta - 1|/0.8$ for $\Delta \in [0.2, 1]$ eV, and $S_{\text{gap}} = 1 - |\Delta - 1.5|/1.5$ for $\Delta \in [1.5, 3]$ eV.

2) Clean-range score ($S_{\text{cleanVB/CBRange}}$): measures the cleanliness of the energy range around the VBM or CBM, defined as $S_{\text{cleanCB/VBRange}} = 1 - |\max(\Delta E_{\text{VB/CB}}, 1) - 1|/0.8$, where $\Delta E_{\text{CB/VB}}$ is the clean range at the VB/CB, truncated at 1 eV because an excessively large energy range is not critical for twisting.

3) DOS score (S_{DOS}): defined as $S_{\text{DOS}} = 1 - D(E_f)/2$, where $D(E_f)$ is the density of states at the Fermi level in states per unit cell per electron volt. A lower density of states results in a higher score.

4) Symmetry score (S_{symmetry}): defined as the number of point group (PG) symmetries divided by 24, which is the maximum number of PG operations in layer groups.

5) Atom number score ($S_{\text{atom number}}$): inversely related to the number of atoms in the unit cell to favor simpler and more symmetrical compounds.

Each component takes value in [0, 1]. The total scores for both semimetals and insulators range from 0 to 1, where higher scores indicate better candidates. Although these scores may involve a degree of arbitrariness, they are based on both theoretical and experimental facts. We intentionally applied less strict criteria to capture a broader range of potentially twistable materials. Although adjusting the definition of the score would alter their rankings, the materials are already categorized into subclasses of lattices, valleys, and SOC strength, and there are relatively few in each. This allows for the manual selection of candidates from each subclass for further study, with the score serving as a reference rather than a strict determinant.

Applying the algorithm

We applied the algorithm to all materials in the 2D-TQCDB (66) and identified 61 candidates as twistable semimetals and 1568 as twistable insulators, with materials further categorized as experimental, computationally exfoliable, computationally stable, and computationally

unstable [see section S5 of the supplementary materials (70) for more details]. Computationally unstable materials are also included to ensure that potential twistable candidates are not overlooked. The significantly smaller number of twistable semimetals compared with insulators is a consequence of the strict requirement for clean, symmetry-protected crossings at the Fermi level, a condition that is difficult to meet despite the relatively large number of 2D topological semimetals (66), which typically exhibit large FSs. A comprehensive list of these materials is provided in sections S6 and S7 of the supplementary materials (70). In the following, we discuss representative twistable semimetals and insulators across different classes.

Representative twistable semimetals

Twistable semimetals have symmetry-protected crossing points near the Fermi level, which can be gapped in the presence of SOC. Table 1 presents representative twistable semimetals, from which three examples are selected and discussed in the following.

Figure 3A shows the monolayer band structure of a predicted stable material $\text{Ta}_2\text{Te}_2\text{S}$ in LG 72 ($p\bar{3}m1$), which features a linear Dirac crossing at the K point without SOC. This crossing is gapped by SOC, resulting in a sizable gap of ~ 100 meV, much larger than the negligible SOC gap in graphene. The large SOC in $\text{Ta}_2\text{Te}_2\text{S}$ leads to the formation of quantum spin Hall (QSH) states in both the valence and conduction bands, which are absent in graphene. After twisting, the moiré band structure is expected to be markedly different from that of graphene and to show distinct moiré physics.

Figure 3B is for the computationally exfoliable material ZrBr in LG 72 ($p\bar{3}m1$), where a quadratic crossing is observed at Γ with an SOC gap of ~ 50 meV. The quadratic dispersion arises from the higher symmetry of the D_{3d} point group at Γ , which includes three in-plane C_2 rotations and the TRS in addition to the C_{3z} rotation. The effective $\mathbf{k} \cdot \mathbf{p}$ Hamiltonian for the quadratic dispersion at Γ has the form

$$h(\delta\mathbf{k}) \approx v_1 \delta\mathbf{k}^2 \sigma_0 + v_2 \begin{bmatrix} 0 & e^{i\frac{\pi}{3}} \delta k_+^2 \\ e^{-i\frac{\pi}{3}} \delta k_-^2 & 0 \end{bmatrix} \quad (3)$$

where $\delta k_{\pm} = \delta k_x \pm i\delta k_y$, and $v_{1,2}$ are the parameters for the two terms. The quadratic dispersion leads to a quasi-flat segment of band near Γ and a large DOS at the Fermi level, which can result in flatter moiré bands after twisting. These flatter bands reduce the kinetic energy of electrons, enhancing correlation effects. Therefore, ZrBr could exhibit strong interactions in its twisted form.

Figure 3C shows the experimental material MoS_2 (71) in LG 15 ($p2_1/m1$), featuring a linear crossing along the Γ -Y high-symmetry line. MoS_2 adopts the $17''$ structure, similar to $17''\text{-WTe}_2$ (72–75). Note that $17''\text{-MoS}_2$ is a metastable phase and is different from the common 2H-MoS_2 (76). In $17''\text{-MoS}_2$, the linear crossing along the Γ -Y line without SOC is protected by a C_2 rotation symmetry. When SOC is introduced, this crossing point is gapped by ~ 50 meV, and both the valence and conduction bands exhibit QSH states. Unlike WTe_2 , in which small Fermi surface pockets remain even with SOC, MoS_2 has a fully gapped band structure, potentially leading to much simpler

moiré band structures and offering a cleaner platform for exploring correlated moiré phases. The WS_2 in the same group exhibits similar gapped band structures and can be synthesized in this structure starting from K_xWS_2 (77).

Among all twistable semimetals, only four candidates have a square lattice: CuCl_2 , PtS , SnS , and RuCl_2 . Of these, CuCl_2 is computationally exfoliable and has a twisting point at the X point, which is gapped by SOC with small magnetic moments developed on Cu. The others, however, are computationally unstable. For the hexagonal lattice, we note that the M valley has only 1D irreducible representations in the absence of SOC and therefore cannot host twistable semimetals.

Representative twistable insulators

Table 2 presents representative twistable insulators. From this list, we select six compounds with distinct properties, with their band structures shown in Fig. 3.

The first three compounds, BiTeI , Sc_2CCl_2 , and ZrNCl , all have a hexagonal lattice but have different valleys and SOC splitting strength.

Figure 3D shows the experimental material BiTeI in LG 69 ($p\bar{3}m1$), with the CBM near the Γ point, characterized by strong Rashba-type SOC splitting. To model the spin splitting, we consider a $\mathbf{k} \cdot \mathbf{p}$ Hamiltonian at Γ point with the first- and second-order terms

$$h(\delta\mathbf{k}) \approx v\delta\mathbf{k} \cdot \boldsymbol{\sigma} + \frac{\delta k_x^2 + \delta k_y^2}{2m} \sigma_0 \quad (4)$$

where v characterizes the strength of the linear SOC term and m is the effective mass. As the Γ valley has the TRS, the strong SOC effects could potentially induce QSH states in the moiré bands. At small twist angles where interactions become dominant, this nontrivial topology may give rise to fascinating correlated physical phenomena, including fractionalized topological states at noninteger fillings.

Figure 3E shows ZrNCl in LG 72 ($p\bar{3}m1$), with the CBM at the K valley and the VBM at the Γ valley, both of which exhibit negligible SOC splitting. The weak SOC splitting at the K valley contrasts sharply with that of the well-studied TMD materials such as MoTe_2 and WSe_2 . In the literature, bulk ZrNCl and its Hf counterpart, HfNCl , have been shown to exhibit superconductivity at ~ 15 K and 25 K, respectively, upon doping into the conduction band (78, 79). Therefore, twisted ZrNCl may similarly exhibit superconducting properties upon doping.

Figure 3F features a predicted exfoliable material, Sc_2CCl_2 in LG 72 ($p\bar{3}m1$), displaying a CBM at the M point. In contrast to the K valley in TMDs, there are three C_3 -related M valleys in the monolayer BZ. When mapped onto the moiré BZ, these M valleys form a kagome lattice in momentum space, which could lead to new correlated physics enriched by the rich valley degrees of freedom (69).

The next two compounds, Cu_2WS_4 and GeS_2 , both have a square lattice. Although moiré systems with a square lattice have been theoretically proposed (61, 80, 81), they have not yet been realized in experiments. Here, we propose two twistable square lattice insulators, Cu_2WS_4 and GeS_2 , both of which are computationally exfoliable (55, 56) with bulk structures and exhibit different valley properties.

Figure 3G shows Cu_2WS_4 in LG 57 ($p\bar{4}2m$), featuring a VBM at the M point with negligible SOC splitting, and a flat CBM in the vicinity of Γ with strong SOC splitting. Both the Γ and M valleys exhibit C_4 symmetry and form a square lattice in the moiré BZ upon twisting. Figure 3H presents GeS_2 in LG 59 ($p\bar{4}2m$), with the VBM at the X point and the CBM at the Γ point. The X point differs from the Γ and M points because it lacks C_4 symmetry. Upon twisting, the X valley forms two sets of staggered square lattices in the moiré BZ, related by the C_4 symmetry, as shown in Fig. 1D.

We note that the Γ , M, and X valleys from the square lattice all respect TRS and could potentially give rise to QSH states in the presence of strong SOC. Moiré systems with a square lattice could serve as a platform for simulating the Hubbard model, particularly in relation

Table 1. Representative twistable semimetals. Semimetals are classified into different classes according to Bravais lattices, valleys, and SOC strengths.

Lattice	SOC	Γ	K	nHSP
Hexagonal	Strong	ZrBr	$\text{Ta}_2\text{Te}_2\text{S}$	ZrTe
	Weak	Cu_2Se	Ge	IrPS_3
		HSP	nHSP	
Rectangular	Strong	–	MoS_2	
	Weak	–	Hg_3S_2	

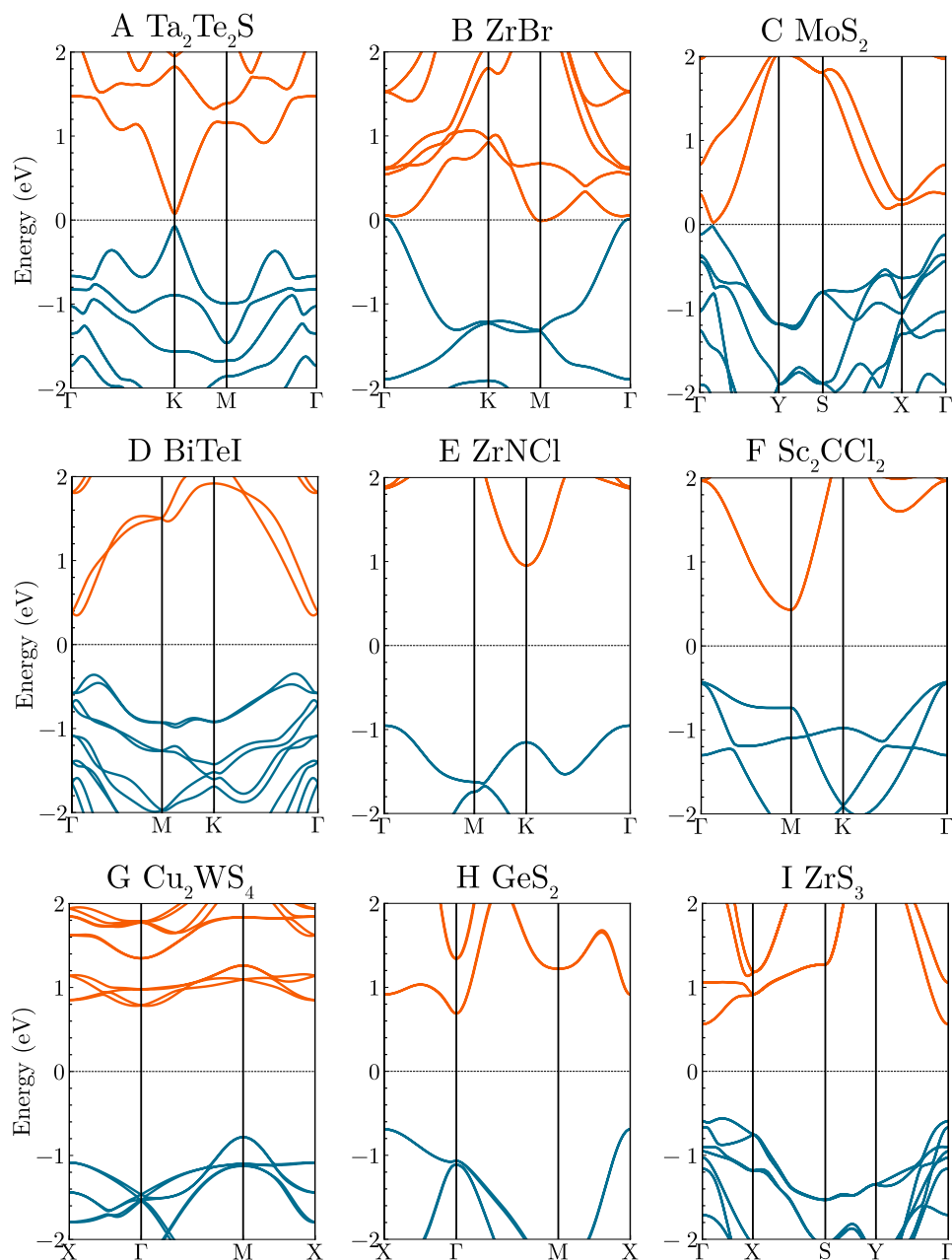


Fig. 3. Monolayer band structures of representative twistable materials. (A) Hexagonal $\text{Ta}_2\text{Te}_2\text{S}$ exhibits a Dirac cone at the K point, which is gapped by SOC with a gap of ~ 100 meV, placing it in a distinctly different universal class from TBG. (B) Hexagonal ZrBr features a quadratic crossing at the Γ point, further gapped by SOC with a gap of ~ 50 meV. (C) Rectangular MoS_2 displaying a linear crossing along the Γ -Y line, which opens a SOC gap of ~ 50 meV. (D) Hexagonal BiTeI hosts the CBM at the Γ point with strong Rashba SOC splitting. (E) Hexagonal ZrNCl features a clean K valley at the CBM with negligible SOC splitting. (F) Hexagonal Sc_2CCl_2 has the CBM at the M valley with negligible SOC splitting. (G) Square Cu_2WS_4 has the VBM at the M valley with negligible SOC splitting. (H) Square GeS_2 has the VBM at the X valley with negligible SOC splitting. (I) Rectangular ZrS_3 hosts the CBM at the Γ valley with negligible SOC splitting.

to high-temperature superconductivity in cuprates (82, 83), and may also exhibit unconventional superconductivity.

Finally, Fig. 3I shows the band structure of ZrS_3 in LG 46 ($pmmn$), which has a rectangular lattice and is computationally exfoliable (55). The CBM appears at Γ and VBM is located along the Γ -X line. In the rectangular lattice, the two in-plane lattice constants differ, and this disparity is further amplified in the moiré unit cell at small twist angles (32, 84). As a result, rectangular moiré systems are expected to exhibit quasi-1D characteristics, potentially acting as Luttinger liquid

simulators (58). In ZrS_3 , the two in-plane lattice constants have a ratio of ~ 1.5 . Additionally, the bands near the VBM are relatively flat along the k_x direction, which could potentially lead to flat moiré bands and further enhance the potential for strongly correlated physics.

Band structures for twisted bilayer materials

The database of theoretically twistable 2D materials developed in this work identifies promising candidates for twist engineering from monolayer properties. These candidates have clean band structures, making them suitable for simple continuum modeling and likely to yield clean moiré bands after twisting. We demonstrate the predictive power of this approach by comparing database-based expectations with full density-functional calculations of selected twisted bilayers, including structural relaxation. Because small twist angles lead to moiré cells with many thousands of atoms, fully ab initio calculations are computationally demanding. We therefore study representative candidates at moderate twist angles. AA-stacked bilayers are constructed by directly aligning the two layers, whereas AB-stacked bilayers are obtained by rotating the top layer by 180° before stacking.

We first focus on the universality class of hexagonal-lattice M-valley with weak SOC. Figure 4, A to E, summarizes the band structures at twist angle $\theta = 7.34^\circ$ for AA-stacked SnSe_2 , SnS_2 , ZrS_2 , HfS_2 , and Sc_2CCl_2 , respectively. All five materials demonstrate the emergence of isolated (or nearly isolated) moiré flat bands, confirming the utility of our twistable materials database. The insets complement these results with moiré bands obtained at a smaller twist angle of 3.89° (69), for which we have already built continuum models. This comparison illustrates that although twist angle variation enables bandwidth control of the few-flat-band physics, the essential characteristics are already established at moderate angles.

We next extend our analysis of moiré bands to additional universality classes, highlighting distinct electronic properties emerging from different valley configurations and SOC strengths. Figure 4F depicts the conduction bands of twisted AB-stacked GaTe , a hexagonal lattice material with the valley at the M point, showing pronounced spin splitting in its moiré bands due to the strong Rashba SOC in the monolayer, offering potential for spin-dependent phenomena. Figure 4, G and H, present the valence bands of AB-stacked SnSe_2 and conduction bands of AB-stacked BiTeI , respectively. These hexagonal-lattice Γ -valley systems with strong SOC provide a promising platform with which to host nontrivial QSH topology, making them promising candidates for topological moiré physics, including fractionalized topological

Table 2. Representative twistable insulators. Insulators are classified into different types of lattices, valleys, and SOC strengths.

Lattice	SOC	Γ	M	K	nHSP
Hexagonal	Strong	BiTeI	GaTe	GaSe	AsSb
	Weak	InSe	Sc ₂ CCl ₂	ZrNCI	PtSe ₂
		Γ	M	X	nHSP
Square	Strong	BiIO	HgH ₂	—	SnI ₂
	Weak	GeCl ₂	Cu ₂ WS ₄	GeS ₂	Gel
		HSP	nHSP		
Rectangular/ oblique	Strong	Sb ₂ Te ₂ O	SnSe		
	Weak	ZrS ₃	SnPS ₃		

states. Figure 4I presents the conduction bands of GeS₂, a square-lattice material with a Γ valley and a weak SOC. The two lowest spinful conduction bands are isolated from other bands, which could be flatter at a lower twisting angle and modeled by a square-lattice Hubbard model. Finally, Fig. 4J shows the moiré bands of twistable semimetal Ta₂Te₂S, originating from the monolayer Dirac cone with a large SOC gap. Compared with TBG, several distinct features emerge in Ta₂Te₂S: Strong SOC induces a substantial gap between the conduction and valence bands, and significant spin splitting reduces the 4-fold valley-spin degeneracy in TBG to 2-fold valley-only degeneracy in Ta₂Te₂S. Note that the densely packed valence bands below -0.2 eV originate from other valleys at nHSP, as seen in the monolayer bands in Fig. 3A.

These diverse case studies of moiré band structures highlight the predictive power of our database. By leveraging monolayer descriptors alone, we can efficiently identify materials poised to host clean, twist-induced quasi-flat bands. In section S6 of the supplementary materials (70), we provide the complete list of theoretically twistable materials in each universality class.

Simple moiré continuum model

Different $\mathbf{Q}$ -lattices constrained by the intravalley symmetries produce different moiré potentials, which control both the single-particle bands and the interaction-driven many-body physics. Below, we present simple continuum models for several typical cases and summarize their main features, including quasi-1D dispersion and nontrivial band topology.

We start with the M valley on the hexagonal lattice with negligible SOC. The M valley of the hexagonal lattice has three C_3 -related subvalleys, forming a kagome $\mathbf{Q}$ lattice, as shown in Fig. 1B. In valley $\eta = 0$, a simplified moiré Hamiltonian has the form

$$\begin{aligned} [h_{\mathbf{Q}\mathbf{Q}'}(\mathbf{k})]_{sl,s'l'} &= \delta_{\mathbf{Q}\mathbf{Q}'} \delta_{ss'} \delta_{ll'} \left[\frac{(k_x - Q_x)^2}{2m_x} + \frac{(k_y - Q_y)^2}{2m_y} \right] \\ &+ [T_{\mathbf{Q}\mathbf{Q}'}]_{sl,s'l'}, \text{ for } \mathbf{Q}^{(\eta)} \in \mathcal{Q}_{l^{(\eta)}}, \end{aligned} \quad (5)$$

where m_x and m_y are two effective masses along the k_x and k_y directions, respectively, and $s(l)$ is the spin (layer) index. The interlayer hopping terms have the form

$$\begin{aligned} [T_{\mathbf{Q}\mathbf{Q}'}^{\text{AA}}]_{ls,(-l)s} &= (\pm i w_1^{\text{AA}} + w_2^{\text{AA}}) \delta_{\mathbf{Q}\pm\mathbf{q}_0, \mathbf{Q}'} + w_3^{\text{AA}} \delta_{\mathbf{Q}\pm(\mathbf{q}_1-\mathbf{q}_2), \mathbf{Q}'} \\ [T_{\mathbf{Q}\mathbf{Q}'}^{\text{AB}}]_{ls,(-l)s} &= w_2^{\text{AB}} \delta_{\mathbf{Q}\pm\mathbf{q}_0, \mathbf{Q}'} + w_4^{\text{AB}} \delta_{\mathbf{Q}\pm(\mathbf{q}_1-\mathbf{q}_2), \mathbf{Q}'} \end{aligned} \quad (6)$$

where $\mathbf{q}_i = C_{3z}^i \frac{\mathbf{b}_{M_1}}{2}$ is the first harmonic vectors for the Fourier expansion of moiré potential in M-valley systems. The superscript AA (AB) is for AA (AB) stacked bilayer, which has space group 149 (150) symmetry.

Because of the negligible SOC, the model presents spin SU(2) symmetry. In (69), we used this simplified M-valley model to fit the moiré bands of SnSe₂ and ZrS₂, which gives good agreement in both dispersion and wavefunction. One key feature of this simplified M-valley moiré Hamiltonian is the emergence of an effective $\tilde{M}_z$ symmetry, which acts nonsymmorphically in momentum space by mapping $\mathbf{k}$ to $\mathbf{k} + \mathbf{q}_0$. The momentum-space nonsymmorphic $\tilde{M}_z$ symmetry, which exists without applied magnetic fields, enforces perfect nesting at wave vector $\mathbf{q}_0$ and induces an effective 1D character (69). Although the three M valleys remain nearly decoupled at the single-particle level, strong electron-electron interactions arising from the flat moiré bands can couple them. Combining with the underlying C_3 symmetry relating the three valleys, an exotic crossed sliding Luttinger liquid phase may be stabilized. These unique characteristics make M-valley moiré systems promising platforms for realizing and studying correlated Luttinger liquid physics in 2D materials.

We now discuss a continuum model of the Γ valley in hexagonal lattice systems. Under the symmetry constraints of C_{3z} and C_{2x} (for AA stacking) or C_{2y} (for AB stacking) and $\mathcal{T}$, the simplified general model has the form

$$\begin{aligned} [h_{\mathbf{Q}\mathbf{Q}'}(\mathbf{k})]_{sl,s'l'} &= \delta_{\mathbf{Q}\mathbf{Q}'} \delta_{ss'} \left[\delta_{ll'} \frac{(\mathbf{k} - \mathbf{Q})^2}{2m_1} + \delta_{-ll'} \frac{(\mathbf{k} - \mathbf{Q})^2}{2m_2} \right] \\ &+ [T_{\mathbf{Q}\mathbf{Q}'}]_{sl,s'l'}. \end{aligned} \quad (7)$$

The specific form of $[T_{\mathbf{Q}\mathbf{Q}'}^{\text{AA}}]$ is provided in section S2 of the supplementary materials (70). Because the top and bottom layer share the same hexagonal $\mathbf{Q}$ lattice, the layer coupling is particularly strong in the Γ valley, making layer bonding and antibonding states the natural bases. To account for the potentially different effective masses of these states, we introduce two mass parameters, m_1 and m_2 . To realize QSH states, strong SOC terms are required in the continuum model, which can be introduced through either the second harmonic SOC term or the on-site Rashba term, depending on the ab initio band structures of the system.

Finally, we examine the continuum model for a general square lattice Γ -valley system with C_{4z} , C_{2y} and $\mathcal{T}$ symmetries. Although the Hamiltonian retains the general form of Eq. 7, the distinct $\mathbf{Q}$ -lattice geometry produces characteristically different moiré potentials and resulting band structures. When the SOC is negligible, we simplify the model by retaining only the onsite term for interlayer coupling $[T_{\mathbf{Q}\mathbf{Q}'}]_{ls,(-l)s} = w_0 \delta_{\mathbf{Q}\mathbf{Q}'}$ and keep the first harmonic terms for the intralayer part, i.e., $[T_{\mathbf{Q}\mathbf{Q}'}]_{ls,ls} = V_1 \delta_{\mathbf{Q}\pm\mathbf{g}_1, \mathbf{Q}'}$. Here, $\mathbf{g}_{1,i} = C_{4z}^i \mathbf{b}_{M_1}$ represents the first harmonic vectors. When the lowest set of moiré bands is topologically trivial, it can be described by a single-orbital square-lattice Hubbard model, which may provide insight into the cuprate physics.

We also briefly examine the emergence of “magic angles” in moiré systems. For twistable semimetals with a Dirac crossing at the Fermi level, magic angles occur when the Fermi velocity in the moiré bands vanishes at the Dirac point, a condition that can be analytically determined from the monolayer Fermi velocity and interlayer coupling. For twistable insulators, moiré bands generally flatten at smaller angles, although bandwidths may vary nonmonotonically due to the strong moiré potential. The magic angle can be defined as the critical angle where band extrema shift away from the valley momentum in the moiré BZ (43, 85). Our 2D-TQCDB database enables preliminary estimates through the band-edge effective masses, where larger masses generally indicate flatter moiré bands and potentially larger magic angles. However, precise predictions require material-specific modeling of the full moiré Hamiltonian, including interlayer coupling strength, which is an important direction for future study.

Experimental validation

A central goal of this work is to identify twistable materials suitable for immediate experimental studies. Because moiré devices are typically

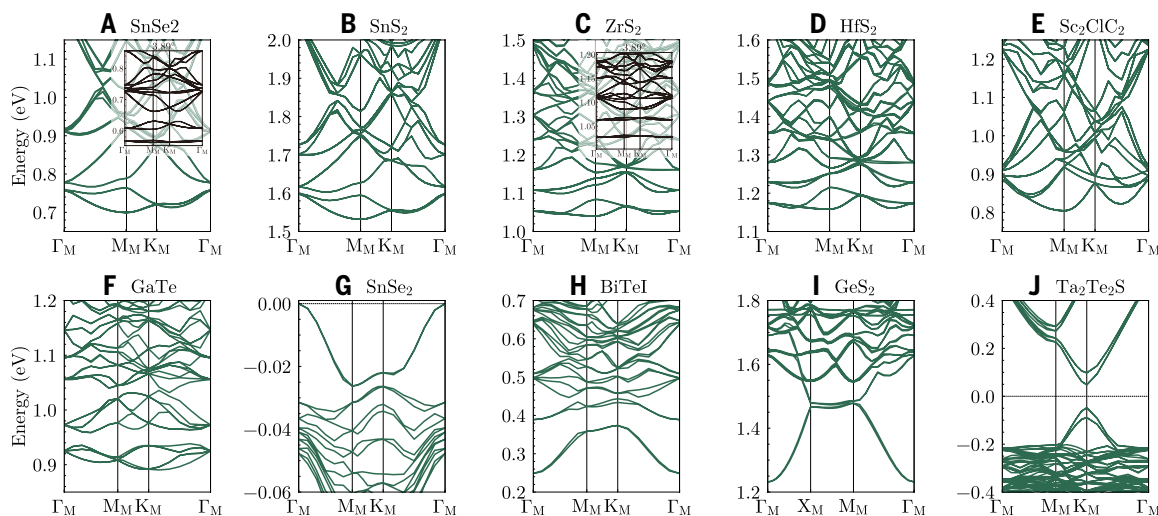


Fig. 4. Band structures for twisted bilayer materials. Shown are conduction bands of SnSe₂ (A), SnS₂ (B), ZrS₂ (C), HfS₂ (D), and Sc₂ClC₂ (E) at twist angle $\theta = 7.34^\circ$, representing hexagonal lattice M-valley systems with negligible SOC. In the inset of (A) and (C), we show the conduction band of SnSe₂ and ZrS₂ at $\theta = 3.89^\circ$ (69). (F) Conduction bands of GaTe at $\theta = 5.09^\circ$, illustrating a hexagonal lattice M-valley system with strong SOC. (G and H) Valence bands of SnSe₂ (G) and conduction bands of BiTeI at $\theta = 5.09^\circ$ (H), exemplifying hexagonal lattice Γ -valley systems with strong SOC. (I) Conduction bands of GeS₂ at $\theta = 12.68^\circ$, demonstrating the square-lattice Γ -valley type. (J) Ta₂Te₂S at $\theta = 7.34^\circ$, a hexagonal lattice K-valley twistable semimetal resembling TBG but with significantly stronger SOC than TBG. The high-energy entangled valence bands come from other valleys.

fabricated from exfoliated mono- or few-layer crystals (86), we focus on systems that satisfy three practical criteria: (i) the structures have bulk crystal analogs, (ii) bulk crystals can be grown with sufficient size and quality, and (iii) the bulk crystals contain van der Waals gaps or have other viable exfoliation pathways. For these materials, we further consider the following factors: (i) the stability of monolayers to brief air exposure; (ii) how clean the systems are expected to be in terms of defects, impurities, or disorder; and (iii) reports of experimental exfoliation down to monolayers.

Based on these criteria, the experimentally relevant twistable candidates can be grouped into six broad classes as follows:

1) Elements: exfoliation of graphene to monolayers kickstarted the field of 2D electronics (87). However, to this day, we have relatively few experimentally exfoliated elements with which to create moiré structures.

2) TMDs: TMDs have had an overwhelming influence on moiré research. Despite this, only a small subset of transition metals has been used to produce twisted moiré systems experimentally or even theoretically. Materials including SnS₂, HfS₂, and ZrS₂ and their selenium analogs have all been exfoliated down to monolayers (86); however, no twistronic devices with them have been reported to date. What makes them even more interesting is that they are all hexagonal M-valley systems, distinct from the well-studied moiré TMDs, which primarily involve the K valley. This distinction leads to significantly different physics (69).

3) Other chalcogenides: chalcogenides are particularly likely to create van der Waals layered materials due to the chalcogen atoms' lone pairs. It is thus not surprising that many candidate twistable materials belong to this class. Some standout materials here include group IIIA monochalcogenides such as GaSe. These are well established as exfoliable van der Waals systems, but few studies have considered using them in heterostructure or twistronic devices. Group IVB trichalcogenides such as ZrSe₃ and TiS₃ are another promising family here, representing well-established exfoliable systems. Tetradymite-type and related chalcogenides of antimony, bismuth, and group IVA metals comprise the last important family of twistable experimental chalcogenides, including Bi₂Te₃.

4) Halides: similar to chalcogenides, van der Waals halides are fairly common. Not all of them, however, are stable to air or moisture, complicating exfoliation in some cases. The most promising systems according to our methodology include PbI₂, BiI₃, and CdI₂. More details can be found in section S3 of the supplementary materials (70).

5) Mixed pnictides, chalcogenides, and halides: both chalcogenides and halides are especially suitable to produce exfoliable materials. Therefore, it is unsurprising that combinations of halide and chalcogenide ions, or either of these with pnictides, also can produce materials good for exfoliation. Some key examples of materials in this class include ZrNCl, which has been exfoliated to monolayers (88) and is reported to be a superconductor when gated (89), and bismuth chalcogenides such as BiTeI, which likewise form stable monolayers that have thus far been unexplored in moiré research. Thiophosphates such as AgInP₂S₆ (90) have also recently been exfoliated experimentally and can potentially produce clean twistable systems.

6) Other materials: most experimental twistable materials fit into the five classes listed above. GeH is one of the candidates outside of this classification. Several oxides also show promise for twisting theoretically, although experimental work may be somewhat challenging, including TiO₂ and the MXene Ti₂CO₂. Finally, one of the highest twist scores for experimentally obtained materials was found in 2D-GaN. This phase of GaN was recently obtained by encapsulation between sheets of graphene (91). Experimental twisting of this system does not appear possible, but it could become an interesting material in the future if synthetic strategies affording freestanding monolayers can be devised.

Among the experimentally twistable materials, we have successfully synthesized several of the most promising ones in bulk form, including the TMDs SnSe₂, SnS₂, and HfS₂, the chalcogenide GaTe, and the pnictohalide ZrNCl. We also demonstrate the experimental exfoliation of several twistable materials into high-quality monolayers, including SnSe₂, HfSe₂, ZrS₂, and 1T'-WS₂. We used an ultraclean gold-assisted exfoliation technique (92), which enables the deterministic exfoliation of a monolayer in each attempt, ensuring reproducibility and near-unity yield in monolayer production. We apply this technique to several twistable materials, including SnS₂, ZrS₂, HfS₂, and ZrNCl. The

exfoliated monolayer flakes have significantly larger areas compared with conventional Scotch tape exfoliation. The exfoliated monolayer samples can be readily used to make twisted bilayer devices and perform characterizations. As an example, we realized dual-gated devices of homobilayer SnSe_2 , which shows a comparable mobility to MoTe_2 . As a further demonstration, we also fabricated a twisted homobilayer HfS_2 device for millikelvin scanning tunneling microscopy/spectroscopy measurements, confirming the suitability of these exfoliated monolayers for twist-device assembly and local spectroscopy. More details are provided in section S3 of the supplementary materials (70).

Discussion and outlook

This work provides a practical starting point for systematic moiré materials design beyond graphene and TMDs. By identifying theoretically twistable semimetals and insulators with clean, low-energy electronic structures, the resulting database highlights materials for which predictive continuum modeling is feasible and experimentally meaningful. These candidates offer diverse lattice symmetries, valley structures, and SOC strengths, opening routes to correlated states and topological phases in previously unexplored moiré systems. Going forward, the database can guide material synthesis and device fabrication, and experimental outcomes can in turn refine the twistability criteria. Coupling this resource with high-throughput moiré calculation and automated model construction would transform the present screening into an iterative platform for discovering and engineering new moiré quantum matter.

Materials and methods

The electronic properties calculations for the monolayer and twisted bilayer system were performed using density functional theory with the Vienna ab initio software package (VASP) (93). The Perdew-Burke-Ernzerhof exchange-correlation functionals (94) were used. More details are provided in section S4 in the supplementary materials (70).

Bulk crystals of selected twistable candidates were synthesized to assess their experimental feasibility, including SnSe_2 , SnS_2 , HfS_2 , GaTe , and ZrNCl . Among them, SnSe_2 , SnS_2 , and HfS_2 were grown by chemical vapor transport methods, GaTe was grown by a self-flux method, and ZrNCl was grown by two-step solid-state synthesis. Crystal quality, phase purity, and composition were characterized by powder x-ray diffraction. Monolayers of several representative materials, including SnSe_2 , HfSe_2 , ZrS_2 , $1T'$ - WS_2 , SnS_2 , HfS_2 , and ZrNCl , were obtained by mechanical exfoliation onto SiO_2/Si - or Au-coated substrates. For materials with low conventional exfoliation yield, we used gold-assisted exfoliation, which produces large-area, clean monolayers suitable for device fabrication. As demonstrations, we fabricated dual-gated homobilayer SnSe_2 devices and twisted homobilayer HfS_2 devices, confirming the compatibility of these materials with standard moiré device assembly. Further synthesis, exfoliation, characterization, and device fabrication details are provided in section S3 of the supplementary materials (70).

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SUPPLEMENTARY MATERIALS

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2D theoretically twistable material database

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Editor's summary

The field of two-dimensional (2D) materials has experienced explosive growth over the past two decades, roughly in parallel with similarly exciting developments in the field of topological materials. There are a number of 2D materials known to have a topologically nontrivial electronic structure, as well as those that form moiré superlattices when twisted. However, an exhaustive computational search for these properties in 2D materials has not been performed. Petralanda *et al.* adapted methods previously used for 3D materials to characterize topological properties of 2D materials and built an online database. Jiang *et al.* then screened this database for materials that could be predicted theoretically to be “twistable,” providing a guide for future studies of 2D moiré systems. —Jelena Stajic

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